

AI Agent Team for SRE & SysAdmin Automation

Three Technology-Stack Proposals

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AI Agent Team for SRE & SysAdmin Automation

Three concrete tech-stack proposals based on agamm/awesome-ai-sre plus current 2026 market research. All three reuse existing frameworks; none require building agent runtimes from scratch.

Shared Agent Role Model

Use this regardless of which stack you pick.

Role	Primary function	Allowed actions
Duty Engineer	L1 alert triage, paging, comms	Read-only on telemetry; auto-acks; routes
SRE Engineer	Investigation, RCA, hypothesis	Reads observability + runs diag commands
Sys Admin	OS patching, config drift, capacity	Stages patches, opens CRs, executes after approval
Security Engineer	CVE triage, exploitability, mitigation	Proposes patches/WAF rules; HITL gate for prod
Compliance Officer	SOC2/ISO/CIS evidence, policy attestation	Read-only; writes evidence + reports

Role	Primary function	Allowed actions
Change Manager <i>(human-required)</i>	Approves prod changes	The HITL gate
Postmortem Writer	Incident summary, action items	Writes docs only

Architecture pattern: AI proposes -> human approves -> AI executes. A centralised orchestrator (supervisor) consistently outperforms decentralised swarms in 2026 research.

Proposal A — Open-Source Self-Hosted (“the FOSS stack”)

Best for: privacy-sensitive orgs, on-prem deployments, regulated industries.

Layer	Tool	Why
Team mgmt UI	Multica	Linear-style assignee/issue model; agents are teammates. Works with Claude Code, Codex, Gemini, Cursor, etc.
Orchestration	LangGraph	Native HITL pause/resume; durable state; deterministic execution — only major framework with first-class human approval gates
Investigation agent	HolmesGPT (CNCF Sandbox)	ReAct-pattern Kubernetes/cloud investigator; MCP-extensible; bring-your-own-model
K8s diagnostics Runbook execution	K8sGPT (CNCF) StackStorm	Lightweight cluster scanner 6000+ actions, event-driven, used in production by Netflix
Patching	Ansible Lightspeed + OpenAI Aardvark	Lightspeed for OS playbooks; Aardvark for code-level CVE patches
Compliance	Comp AI (open-source)	Continuous endpoint agent for SOC2/CIS evidence; natural-language test generation
Observability	Grafana stack (Loki / Tempo / Mimir) + Grafana Assistant, or SigNoz	Open-source; OTel-native
Protocols	MCP for tools, A2A for inter-agent	Multica + HolmesGPT both speak MCP

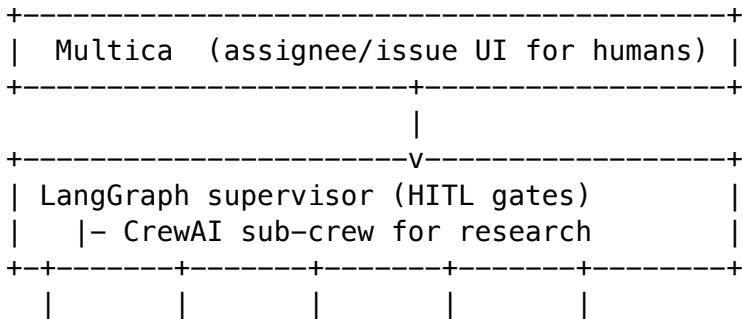
Proposal B — Commercial / Managed (“the enterprise stack”)

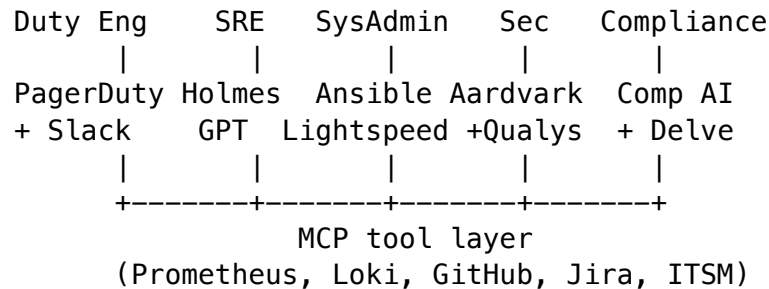
Best for: orgs that want vendor support, faster time-to-value, and have budget.

Layer	Tool	Why
Team mgmt UI	Paperclip	Org-chart + budget governance + per-agent spend caps — better fit when AI labor is treated as a cost center
Orchestration	CrewAI	Role / goal / backstory abstraction maps cleanly to job titles; fastest to assemble; A2A support
SRE Agent	Azure SRE Agent (GA)	Policy guardrails, no-code workflow builder, restart / scale / rollback with mandatory approval
Sys admin / patching	Tenable Hexa AI custom agents	MCP-based orchestration; built for vulnerability remediation at machine speed
Security validation	Qualys Agent Val	TruConfirm exploitability proof + virtual patches / WAF / network containment as bridges to real patches
Compliance	Delve or EasyAudit	Auto-screenshot evidence, pre-mapped controls for SOC2 / ISO27001 / HIPAA
Incident mgmt	incident.io or Rootly	Slack-native, AI postmortems
Observability	Datadog Bits AI or Dynatrace Davis	Mature AIOps; runbook-aware
ChatOps	Claude Code / Codex via Multica or Paperclip workers	Coding role for code-level fixes

Proposal C — Hybrid Pragmatic (“recommended starting point”)

Mix open-source where flexibility matters, commercial where reliability matters.





Why this combo

- **Multica** as the human-facing surface — engineers already think in tickets.
- **LangGraph** as the supervisor (state, persistence, HITL) — solves the “agent sprawl” reliability problem flagged in Datadog’s 2026 *State of AI Engineering*.
- **CrewAI** for the parallel research phase only (cheap, fast multi-perspective brainstorming), then handoff to LangGraph for execution. This is the documented hybrid pattern used in production.
- **HolmesGPT** for live investigation — actively maintained, CNCF-backed, MCP-extensible.
- **Ansible Lightspeed + Aardvark + Qualys Agent Val** covers the patching pipeline end-to-end (code -> OS -> runtime mitigation).
- **Comp AI + Delve** splits continuous monitoring from audit-ready reporting.

Cross-Cutting Concerns

Don’t skip these:

1. **HITL gates are non-negotiable for prod actions.** Bake approval into the LangGraph state machine, not bolt it on.
2. **Runbooks > model selection.** Production case studies (STCLab, others) found runbook quality determined investigation quality far more than which LLM was used.
3. **Budget caps per agent.** Both Paperclip and Multica support this — turn it on day one. AutoGen-style unbounded loops can rack up four-figure bills.
4. **AIBOM + audit trails.** Track which model + framework + tool version was used for every action. SOC2 CC8.1 now requires this for AI-mediated changes.
5. **Avoid AutoGen for new builds.** Microsoft put it in maintenance mode; use the broader Microsoft Agent Framework or one of the above instead.
6. **Start narrow.** Recommended ramp: read-only investigation -> low-risk PR auto-merge for minor CVEs -> reboot / scale with approval -> broader remediation.

Recommendation

Start with **Proposal C**. Specifically:

1. Stand up **Multica** as the front door (Week 1–2).
2. Wire **HolmesGPT + Comp AI** as read-only agents first (Week 2–4).
3. Add **LangGraph supervisor** with HITL approval gates (Week 4–6).

4. Layer **Ansible Lightspeed + Aardvark** for patching with mandatory human approval (Week 6–10).
5. Only then enable autonomous low-risk actions (minor-version PR merges, restart-on-OOM).

This avoids vendor lock-in, leverages CNCF-graded components, and matches the supervisor + specialists architecture that 2026 research consistently shows outperforms decentralised swarms.

Sources

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- [CrewAI vs LangGraph vs AutoGen — DataCamp](#)
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